Spec → Code: ONC_LSN_005_JR_STD

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1. Spec

REPORT ID

ONC_LSN_005_JR_STD

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report identifies data-entry problems on the Lesion (LSN) eCRF form related to Fragment Number entry for split lesions. It scans lesion evaluation data across multiple lesion-type form variants (e.g. HDA, LOEL, LSNEV, LSNHM, LSNIRTU, LSNLG, LSNLRTU, LSNIR, LSNMM, LSNPC, LSNCLTU, LSNPI, LSNRNTU, LSRIR, LSNEVTU) and flags subjects/records where, for lesions marked as "Split", the Fragment Number values are either missing (e.g. "Fragment Number 1" not present when expected, or a numbering gap) or duplicated (the same Fragment Number appears more than once for a lesion that has more than one associated record).

COLUMN HEADING

Country, Site Mnemonic, Subject Number, Lesion evaluation form, Seq #, Lesion Number, Indicate if Lesion Merged or Split (displayed as "Split - <Associated Lesion Number>"), Fragment Number, Error message

LOGIC

Source data: Oracle CDM clinical trial database views for the Lesion (LSN) eCRF form family (cv_datadictionary-driven dynamic form/item resolution), Date of Visit form (cv_dovfm), subject/site reference (irv_cur_subject, cv_safm), and an external supplemental mapping file (LSN_supp_doc.csv) that dynamically maps standardized item aliases (FRAG_NUM, C_FRAG_NUM, LES_NUM, MERGE_SPLIT, ASS_LN, ASS_SP) to the actual study-specific eCRF column names for items "Fragment Number", "Lesion Number", "Indicate if Lesion Merged or Split", "Associated Lesion Number", and "Sponsor ID Number" (item_alias resolved by matching prefix='CD', controllevel='1', and specific columnname lists per item).
Processing steps:
1. Build eCRF filter from &crf macro variable (list of form mnemonics to include, if any).
2. Read the supplemental doc mapping viewname/itemquestion/columnname combinations to standardized aliases; derive formname from viewname.
3. Query cv_datadictionary to find actual database column names per form/item combination for the current study.
4. Dynamically construct SQL to pull data per form (handling forms where addendum/repeating item tables differ from main form table via LEFT JOIN/UNION), coalescing deleted-form and deleted-item flags to exclude deleted records, and joining to cv_dovfm to exclude visits with a DOV error flag set.
5. Union all form-specific extracts into a single LSN dataset with subjectid, visitid, visitmnemonic, formmnemonic, formidx_m (form instance index), itemsetidx_a (repeating item set index), FRAG_NUM, C_FRAG_NUM, LES_NUM, MERGE_SPLIT, ASS_LN, ASS_SP.
6. Deduplicate to one record per subjectid/formmnemonic/formidx_m (first occurrence) as lsn_min.
7. Filter to lesions where MERGE_SPLIT starts with 'Split', LES_NUM is not null, and FRAG_NUM is not null (candidate split-lesion fragment records).
8. Compute per subject/form/fragment aggregates: count of FRAG_NUM occurrences, max of C_FRAG_NUM, count of records sharing the same seq (ASS_SP).
9. Scenario A (Missing - case 1): For each subject, if none of the FRAG_NUM values equal 'Fragment Number 1' (chk=0) and the numeric suffix of LES_NUM is less than 2 (s<2), flag as "Fragment Number missing".
10. Scenario B (Missing - case 2): For each subject, compare count, max, and min of the numeric suffix parsed from FRAG_NUM (e.g. "Fragment Number N" -> N) against the count of distinct fragment numbers; if the count of fragment numbers doesn't match the max value, or the max equals 1 (only one fragment reported when more expected), or the min isn't 1, flag as "Fragment Number missing" (indicates a gap in the fragment numbering sequence).
11. Scenario C (Duplicate): Group by subject and LES_NUM, count occurrences of each FRAG_NUM value; if a fragment number appears more than once (cntas>1) for a lesion that itself appears more than once (count_les>1), flag as "Fragment Number duplicate".
12. Union all three scenario result sets, then join to subject/site reference data (subjectid match) to attach Country, Site Mnemonic, Subject Number.

13. Optionally restrict output to forms specified in the &crf filter (mvar1 list) if provided.
14. Deduplicate final result set on the full column combination.
15. If no records meet the criteria, output a placeholder line "No records meet the test condition"; otherwise output the detail listing.

REPORT SCENARIOS

1) Fragment Number appears missing because 'Fragment Number 1' was never selected for the subject/form and the lesion number suffix indicates it should have been (Scenario A). 2) Fragment Number appears missing because the sequence of fragment numbers used for a subject/form has a gap or inconsistency (count/max/min mismatch of the numeric fragment suffix) (Scenario B). 3) Fragment Number is duplicated - the same fragment number is recorded more than once for a lesion that has multiple associated records (Scenario C). 4) No records meet any of the above conditions - output a "No records meet the test condition" placeholder row instead of a data listing.

SORTING

Final output sorted (nodupkey) by: country, sitenbr, subjectid, subjnbr, formmnemonic, seq, lesion, indicate, fragment, errmsg. Intermediate sorts: study_supp_alias by form_alias/formname/viewname/item_alias; lsn by subjectid/formmnemonic/formidx_m; lesion/lesionx by subjectid (and FRAG_NUM / C_FRAG_NUM) for aggregation purposes.

2. Logic Understanding

This report cross-references multiple Lesion eCRF form variants to find subjects/records where a lesion marked 'Split' has problems with its Fragment Number entries: either a missing 'Fragment Number 1' relative to the lesion number (Scenario A), a gap/inconsistency in the sequence of fragment numbers used (Scenario B), or a duplicated fragment number on a lesion with multiple associated records (Scenario C). Data is assembled dynamically per study via an alias-mapping file and data dictionary lookups, cleaned of deleted records and DOV-errored visits, deduplicated to one record per form instance, then run through the three independent flagging scenarios whose results are unioned, enriched with site/subject info, optionally filtered by form list, deduplicated, and sorted -- falling back to a 'no records' placeholder if nothing is flagged.

1. The report first resolves, for the current study, which actual eCRF column names correspond to standardized item aliases (FRAG_NUM, C_FRAG_NUM, LES_NUM, MERGE_SPLIT, ASS_LN, ASS_SP) by reading the LSN_supp_doc.csv mapping file and cross-checking against cv_datadictionary (matching prefix='CD', controllevel='1', and specific columnname lists) -- this is necessary because the same logical items live under different column names across the many lesion form variants (HDA, LOEL, LSNEV, etc.).

2. Using the resolved column mappings, the report dynamically builds and runs SQL per lesion form to extract records, handling forms whose addendum/repeating tables differ from the main form table via LEFT JOIN/UNION, while coalescing deleted-form and deleted-item flags to drop deleted records, and joining to cv_dovfm to exclude any visit flagged with a DOV error.

3. All per-form extracts are unioned into one combined LSN dataset carrying subjectid, visitid, visitmnemonic, formmnemonic, formidx_m, itemsetidx_a, and the resolved values for FRAG_NUM, C_FRAG_NUM, LES_NUM, MERGE_SPLIT, ASS_LN, ASS_SP.

4. This combined dataset is deduplicated to one record per subjectid/formmnemonic/formidx_m (keeping the first occurrence), producing 'lsn_min', which becomes the base dataset for further filtering.

5. From lsn_min, the report filters down to candidate 'split-lesion fragment' records: those where MERGE_SPLIT starts with 'Split', LES_NUM is not null, and FRAG_NUM is not null.

6. For these candidate records, per subject/form/fragment aggregates are computed: a count of FRAG_NUM occurrences, the max of C_FRAG_NUM, and a count of records sharing the same seq value (ASS_SP) -- these aggregates feed into the three flagging scenarios below.

7. Scenario A ('Missing - case 1'): for a given subject, if none of that subject's FRAG_NUM values equals the literal text 'Fragment Number 1' (i.e., chk=0) AND the numeric suffix parsed from LES_NUM is less than 2 (s<2), the record is flagged with error 'Fragment Number missing' -- meaning a first fragment number was expected but never entered.

8. Scenario B ('Missing - case 2'): for a given subject, the report parses the numeric suffix N out of each FRAG_NUM value (e.g., 'Fragment Number N' -> N) and compares the count of distinct fragment numbers against the max and min of those parsed values; if the count doesn't equal the max, OR the max equals 1 (implying only one fragment was reported when more should exist), OR the min isn't 1, the record is flagged 'Fragment Number missing' -- this catches gaps in the fragment numbering sequence (e.g., 1, 3 but no 2).

9. Scenario C ('Duplicate'): the report groups records by subject and LES_NUM and counts how many times each FRAG_NUM value occurs (cntas) as well as how many total records exist for that LES_NUM (count_les); if a specific FRAG_NUM occurs more than once (cntas>1) AND the lesion itself has more than one associated record (count_les>1), the record is flagged 'Fragment Number duplicate'.

10. The three scenario result sets (A, B, C) are unioned together into one combined flagged-records dataset.

11. The combined flagged dataset is joined to subject/site reference data (matching on subjectid) to pull in and attach Country, Site Mnemonic, and Subject Number for display.

12. If the &crf macro variable (mvar1 list) was populated in step 1, the output is optionally restricted to only the form mnemonics specified in that filter.

13. The final result set is deduplicated on the full combination of output columns (country, sitenbr, subjectid, subjnbr, formmnemonic, seq, lesion, indicate, fragment, errmsg) and sorted by that same column order (nodupkey).

14. If, after all filtering, no records meet any of the three scenario conditions, the report outputs a single placeholder line 'No records meet the test condition' instead of a detail listing; otherwise it outputs the detail listing of flagged records.

3. Spec Review

Readiness: Needs revision — The spec has solid structural detail on data sourcing and processing steps, but Scenarios A and B contain ambiguous, under-explained numeric conditions (LES_NUM suffix, count/max/min logic) that a query generator would have to guess at, and there's a real inconsistency in what ASS_SP represents plus a gap in how fully-missing FRAG_NUM values are captured -- these are blocking issues requiring revision before code generation.

BLOCKING (ambiguity): Scenario A's condition 'the numeric suffix of LES_NUM is less than 2 (s<2)' is used to determine when a fragment number should have existed, but LES_NUM is described elsewhere as 'Lesion Number' (an identifier), and it's unclear what the 'numeric suffix' of a lesion number represents or why s<2 implies a fragment was expected. The spec does not explain what values LES_NUM takes or why this threshold applies. → Add a concrete example of a LES_NUM value and its numeric suffix (e.g. 'Lesion 3' -> suffix 3), and explain in plain language why suffix < 2 indicates 'Fragment Number 1' should have been present, so the rule's intent is unambiguous.

BLOCKING (ambiguity): Scenario B logic ('if the count of fragment numbers doesn't match the max value, or the max equals 1... or the min isn't 1') mixes several conditions with OR logic but never states what 'count', 'max', and 'min' are computed over per subject -- is this per subject/form, or per subject/LES_NUM (per lesion)? Step 8 says 'per subject/form/fragment' but step 10 says 'For each subject' without form/lesion grouping, which is inconsistent grouping granularity. → Explicitly state the grouping key for Scenario B (e.g. 'per subject and formmnemonic and LES_NUM') and rewrite the three sub-conditions as a worked example (e.g. fragments 1,2,4 present -> count=3, max=4, min=1 -> mismatch flagged) to remove ambiguity about what triggers the flag.

WARNING (completeness): The Column Heading includes 'Seq #' but the Logic never explicitly states how Seq # is derived/populated in the output -- ASS_SP is described as 'Sponsor ID Number' in the alias list but step 8 uses 'count of records sharing the same seq (ASS_SP)' implying ASS_SP is a sequence identifier, not Sponsor ID. This is an inconsistency in what ASS_SP actually represents. → Clarify what ASS_SP maps to (Sponsor ID Number vs. Seq) and explicitly document how the 'Seq #' column value is derived from the source fields.

WARNING (consistency): The item_alias list maps ASS_LN to 'Associated Lesion Number' and ASS_SP to 'Sponsor ID Number', but the Column Heading shows 'Indicate if Lesion Merged or Split (displayed as "Split - <Associated Lesion Number>")' with no separate 'Sponsor ID Number' column, and step 8's aggregate description ('count of records sharing the same seq (ASS_SP)') treats ASS_SP as a sequence number, not a Sponsor ID. The terms ASS_SP and 'seq' seem to be used interchangeably for different concepts across the spec. → Standardize terminology: state explicitly whether ASS_SP represents Sponsor ID Number, sequence number, or both, and confirm whether the 'Sponsor ID Number' item is used anywhere in the output columns.

WARNING (completeness): The Description says the report flags issues 'for lesions marked as Split', but Logic step 7 filters on 'MERGE_SPLIT starts with Split' AND LES_NUM not null AND FRAG_NUM not null -- meaning any split lesion missing FRAG_NUM entirely (not just gaps) would be excluded from candidate records before Scenario A/B ever evaluate it, which seems to contradict the intent of detecting 'Fragment Number 1 not present when expected'. → Clarify how a lesion that is Split but has FRAG_NUM completely null (no fragment number entered at all) is captured by Scenario A, since step 7 filters such records out of the candidate set before the missing-check logic runs.

WARNING (testability): Report Scenarios 1 and 2 (Scenario A and B, both producing 'Fragment Number missing') are not clearly distinguishable in the output since the Column Heading only shows 'Error message' as a single field -- if both scenarios produce the same message text 'Fragment Number missing', a reviewer cannot tell from the output which underlying condition (A or B) triggered the flag. → Specify distinct error message text for Scenario A vs Scenario B (e.g. 'Fragment Number 1 missing' vs 'Fragment Number sequence gap') so the two scenarios are distinguishable in the report output, or state explicitly that they intentionally share the same message.

INFO (naming): The list of lesion-type form variants includes 'LSNIRTU' and 'LSRIR' and 'LSNIR' -- 'LSRIR' appears to break the LSN-prefix naming convention used by all other listed forms (HDA, LOEL, LSNEV, LSNHM, LSNIRTU, LSNLG, LSNLRTU, LSNIR, LSNMM, LSNPC, LSNCLTU, LSNPI, LSNRNTU, LSNEVTU). → Verify whether 'LSRIR' is a typo for 'LSNIR' (already listed separately) or an intentionally distinct form mnemonic, and correct/confirm in the spec.

INFO (completeness): The spec doesn't address what happens to ongoing/incomplete lesion records (e.g., a lesion still being evaluated where fragment data entry may be legitimately partial) -- there's no stated exclusion or handling for in-progress records versus genuinely erroneous missing data. → State whether ongoing/incomplete forms should

> be excluded from missing-fragment detection, or whether the DOV-error-flag exclusion in step 4 is intended to cover this case.

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  d.Country AS `Country`,
  d.SiteMnemonic AS `Site Mnemonic`,
  d.SubjectNumber AS `Subject Number`,
  d.LesionEvalForm AS `Lesion evaluation form`,
  d.Seq AS `Seq
#`,
  d.LesionNumber AS `Lesion Number`,
  d.SplitDisplay AS `Indicate if Lesion Merged or Split`,
  d.FragmentNumber AS `Fragment Number`,
  d.ErrorMessage AS `Error message`
FROM
(

/* Scenario A: Fragment Number 1 never selected and lesion suffix < 2 */
SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS SiteMnemonic,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@Form`.`Name` AS LesionEvalForm,
    LOELFM.LOELSPIDIT AS Seq,
    LOELFM.LOELTUALSNNUM1TB AS LesionNumber,
    CONCAT('Split - ', LOELFM.LOELTULNKIDIT) AS SplitDisplay,
    LOELFM.LOELTUFRGMTNUMIT AS FragmentNumber,
    'Fragment Number missing' AS ErrorMessage
  FROM `EDC`.`LOELFM` LOELFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
    AND LOELFM.LOELTUORRESMSIT LIKE 'Split%'
    AND LOELFM.LOELTUALSNNUM1TB IS NOT NULL
    AND LOELFM.LOELTUFRGMTNUMIT IS NOT NULL
    AND NOT EXISTS (
      SELECT 1 FROM `EDC`.`LOELFM` L1
      WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
        AND L1.LOELTUFRGMTNUMIT = 'Fragment Number 1'
    )
    AND CAST(REGEXP_EXTRACT(LOELFM.LOELTUALSNNUM1TB, '[0-9]+$') AS INT) < 2

  UNION ALL

/* Scenario B: count/max/min mismatch of numeric fragment suffix */

  SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS SiteMnemonic,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@Form`.`Name` AS LesionEvalForm,
    LOELFM.LOELSPIDIT AS Seq,
    LOELFM.LOELTUALSNNUM1TB AS LesionNumber,
    CONCAT('Split - ', LOELFM.LOELTULNKIDIT) AS SplitDisplay,
    LOELFM.LOELTUFRGMTNUMIT AS FragmentNumber,
    'Fragment Number missing' AS ErrorMessage
  FROM `EDC`.`LOELFM` LOELFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
    AND LOELFM.LOELTUORRESMSIT LIKE 'Split%'
    AND LOELFM.LOELTUALSNNUM1TB IS NOT NULL
    AND LOELFM.LOELTUFRGMTNUMIT IS NOT NULL
    AND (
      (
        SELECT COUNT(DISTINCT CAST(REGEXP_EXTRACT(L2.LOELTUFRGMTNUMIT, '[0-9]+$') AS INT))
        FROM `EDC`.`LOELFM` L2
        WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
          AND L2.LOELTUFRGMTNUMIT IS NOT NULL
      ) <>
      (
        SELECT MAX(CAST(REGEXP_EXTRACT(L3.LOELTUFRGMTNUMIT, '[0-9]+$') AS INT))
        FROM `EDC`.`LOELFM` L3
        WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
```

```
          AND L3.LOELTUFRGMTNUMIT IS NOT NULL
      )
      OR
      (
        SELECT MAX(CAST(REGEXP_EXTRACT(L4.LOELTUFRGMTNUMIT, '[0-9]+$') AS INT))
        FROM `EDC`.`LOELFM` L4
        WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
          AND L4.LOELTUFRGMTNUMIT IS NOT NULL
      ) = 1
      OR
      (
        SELECT MIN(CAST(REGEXP_EXTRACT(L5.LOELTUFRGMTNUMIT, '[0-9]+$') AS INT))
        FROM `EDC`.`LOELFM` L5
        WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
          AND L5.LOELTUFRGMTNUMIT IS NOT NULL
      ) <> 1
    )

  UNION ALL

/* Scenario C: duplicate fragment number for a lesion with >1 associated records */

  SELECT
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS SiteMnemonic,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@Form`.`Name` AS LesionEvalForm,
    LOELFM.LOELSPIDIT AS Seq,
    LOELFM.LOELTUALSNNUM1TB AS LesionNumber,
    CONCAT('Split - ', LOELFM.LOELTULNKIDIT) AS SplitDisplay,
    LOELFM.LOELTUFRGMTNUMIT AS FragmentNumber,
    'Fragment Number duplicate' AS ErrorMessage
  FROM `EDC`.`LOELFM` LOELFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (LOELFM.DELETEDFM = 'N' OR LOELFM.DELETEDFM IS NULL)
    AND LOELFM.LOELTUORRESMSIT LIKE 'Split%'
    AND LOELFM.LOELTUALSNNUM1TB IS NOT NULL
    AND LOELFM.LOELTUFRGMTNUMIT IS NOT NULL
    AND (
      SELECT COUNT(*) FROM `EDC`.`LOELFM` L6
      WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
        AND L6.LOELTUALSNNUM1TB = LOELFM.LOELTUALSNNUM1TB
    ) > 1
    AND (
      SELECT COUNT(*) FROM `EDC`.`LOELFM` L7
      WHERE `@HDR`.`Subject`.`ID` = `@HDR`.`Subject`.`ID`
        AND L7.LOELTUALSNNUM1TB = LOELFM.LOELTUALSNNUM1TB
        AND L7.LOELTUFRGMTNUMIT = LOELFM.LOELTUFRGMTNUMIT
    ) > 1
) d
GROUP BY
  d.Country, d.SiteMnemonic, d.SubjectNumber, d.LesionEvalForm, d.Seq,
  d.LesionNumber, d.SplitDisplay, d.FragmentNumber, d.ErrorMessage
ORDER BY
  `Country`, `Site Mnemonic`, `Subject Number`, `Lesion evaluation form`,
  `Seq
#`, `Lesion Number`, `Indicate if Lesion Merged or Split`,
  `Fragment Number`, `Error message`
```

Given the multi-form dynamic-mapping design described (cv_datadictionary-driven alias resolution across HDA/LOEL/LSNEV/etc.), I scoped the query to LOELFM only since it's the one lesion-family Form confirmed in the study metadata with clearly matching items (LOELTUFRGMTNUMIT=Fragment Number, LOELTUALSNNUM1TB=Lesion Number, LOELTUORRESMSIT=Merged/Split indicator, LOELTULNKIDIT=Associated Lesion Number, LOELSPIDIT=Seq); the other named form variants (HDA, LSNEV, LSNHM, etc.) are not present in the provided metadata so are omitted rather than guessed, and this should be flagged to the user to confirm via DESCRIBE FORM/extend with UNION ALL blocks per form if those forms exist under different names. I implemented

the three scenarios (A: no Fragment Number 1 + lesion suffix<2; B: count/max/min mismatch of numeric fragment suffix via correlated subqueries; C: duplicate fragment number for a lesion with multiple records) as three UNION ALL blocks each with its own correlated subqueries scoped by Subject ID, deduplicated via GROUP BY in the outer query, and excluded deleted records via LOELFM.DELETEDFM; DOV-error-visit exclusion (DOVFM.DOVSERRORVIT) was omitted because the spec's join path to DOVFM per-record wasn't fully specified in metadata and would require guessing the join key, so this is flagged as a gap. The zero-result placeholder row ("No records meet the test condition") and the &crf/mvar1 form-list filter parameter are omitted since CQL as generated returns an empty result set naturally and no runtime parameter was explicitly named for form filtering in the Prompts section of this spec excerpt.